Kansinee Khuttiya

Seeking Software Engineering Internship · Available immediately

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EDUCATION

Assumption University (ABAC) | B.Eng. Computer Engineering *Expected Graduation, October 2027*

Focus: AI engineering & cybersecurity. Relevant coursework: Data Mining, Machine Learning, Computer Networks, Operating Systems, Parallel & Distributed Computing.

PROJECTS

NYXUS — Security-First RAG Chatbot *Full-stack · solo project*

github.com/evieffvy/NYXUS · nyxus-phi.vercel.app

- Architected end-to-end **RAG application** (Next.js 15 + FastAPI + Postgres + Gemini): PDF ingestion, fixed-size overlap chunking, gemini-embedding-001 with cosine top-k retrieval and citation injection; NextAuth v5 (Google OAuth, JWT, bcrypt)
- Implemented **prompt-injection defense** across 7 weighted signal classes (instruction override, role hijack, jailbreaks, delimiter smuggling); blocks score ≥ 0.6 before reaching the LLM
- Built **PII redaction engine** (10+ pattern classes incl. Luhn-validated cards, AWS keys, GitHub PATs, JWTs) and **OWASP Top 10 code scanner** with structured-output Gemini and severity-rated findings
- Engineered append-only **audit log**, hardened security headers (CSP, HSTS), per-IP rate limiting (slowapi); GitHub Actions CI on every push
- Tech: Next.js 15, React 19, TypeScript, Tailwind 4, NextAuth v5, FastAPI, Pydantic, slowapi, Prisma, Postgres (Supabase), Gemini 2.0 Flash

HORUS — AI-Powered Threat Intelligence Dashboard *Full-stack · solo project*

github.com/evieffvy/HORUS · horus-vulnscope.vercel.app

- Built **CVE intelligence dashboard** ingesting live data from **NVD REST API v2.0**; CVSS v3.1/v3.0/v2 normalization with severity-band mapping (CRITICAL ≥ 9.0, HIGH ≥ 7.0)
- Integrated **Gemini 1.5 Flash** for structured Thai-language CVE summaries (affected systems, attacker capabilities, urgency); streaming bilingual chat with active-CVE context injection
- Filterable analyst UI (keyword / severity / date / CVSS), Recharts severity distribution, jsPDF export of filtered reports for offline triage; rate-limit-aware NVD fetching
- Tech: Next.js 14, TypeScript, Tailwind, Google Generative AI SDK, Recharts, jsPDF, NVD API

SYCL Parallel Computing — NIST SP 800-22 Frequency Tests *Systems · C++17 · Intel oneAPI*

github.com/evieffvy/sycl_project · evieffvy.github.io/sycl_project

- Reimplemented **NIST SP 800-22** Monobit and Block Frequency tests as SYCL parallel kernels, achieving **60× GPU speedup** over the NIST STS serial baseline at 1 GB input (32 Gbit/s sustained throughput)
- Monobit: two-phase work-group reduction with **sycl::atomic_ref**; Block Frequency: two-pass data-parallel design with **sycl::reduction** in double precision for chi-squared accumulation
- Single portable codebase targeting Intel CPU, Intel GPU, NVIDIA GPU, and AMD GPU via SYCL/DPC++ — no CUDA vendor lock-in; P-values match NIST STS 2.1.2 within 1×10^{-5}
- Tech: C++17, SYCL/DPC++, Intel oneAPI, CMake, Linux

ML Coursework · Personal Portfolio *Python · scikit-learn*

- **ML coursework** — CNN transfer learning (Stanford Dogs, 120 classes) · PCA eigenface recognition (NumPy/OpenCV) · IMDB sentiment (LR / Naive Bayes / TF-IDF) · Apriori market basket · Iris K-Means (elbow + silhouette) · Titanic ID3 — github.com/evieffvy
- **Portfolio** (evieffvy.vercel.app) — Next.js + Tailwind + Framer Motion

TECHNICAL SKILLS

- **Languages:** Python, C, C++, TypeScript, JavaScript, SQL, Bash, HTML, CSS
- **AI / ML:** scikit-learn, OpenCV, CNN transfer learning, PCA, K-Means, decision trees (ID3), Apriori, RAG, embeddings, LLM integration (Google Gemini), streaming AI chat
- **Web:** Next.js 15, React 19, Tailwind CSS, Framer Motion, NextAuth, FastAPI, Pydantic, REST, SSE streaming, Recharts, jsPDF
- **Security:** OWASP Top 10, prompt-injection defense, PII redaction, CVE / NVD analysis, audit logging, security headers (CSP, HSTS), rate limiting, JWT / OAuth
- **Data & Infra:** Postgres, Prisma, Supabase, Linux, Git, CMake, SYCL / oneAPI, Vercel, Render

LANGUAGES

Thai (Native) · **English** (Fluent) · **Mandarin** (Learning, HSK 3)